

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 2(a + 3b) =$$

$$(2) \quad 5(4x + 7) =$$

$$(3) \quad 2(4x - 3) =$$

$$(4) \quad 4(2y + 1) =$$

$$(5) \quad 4(3a + 1x) =$$

$$(6) \quad 5(a + 2y) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4(b - 2) =$$

$$(2) \quad -2(x + 1a) =$$

$$(3) \quad 3(2y + 1) =$$

$$(4) \quad 3(a + 6) =$$

$$(5) \quad 5(2x + 3y) =$$

$$(6) \quad 4(y - 4) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 2(b + 2x) =$$

$$(2) \quad -2(x + 1y) =$$

$$(3) \quad -4(3a - 4b) =$$

$$(4) \quad -5(x - 1a) =$$

$$(5) \quad 5(a - 6) =$$

$$(6) \quad 5(y + 4) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 5(2x + 2) =$$

$$(2) \quad 2(3y - 4b) =$$

$$(3) \quad 4(2a + 2) =$$

$$(4) \quad 2(3a - 1) =$$

$$(5) \quad 5(2x + 5) =$$

$$(6) \quad 2(x - 1b) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4(x - 3a) =$$

$$(2) \quad 2(3b + 8) =$$

$$(3) \quad 4(4b + 4) =$$

$$(4) \quad 4(2a + 4) =$$

$$(5) \quad 3(3a + 2) =$$

$$(6) \quad 4(y - 3a) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4x + 4(4x + 1y) =$$

$$(2) \quad 4(x + 2) + 3(x - 5) =$$

$$(3) \quad 4(y + 3) + 3(y - 5) =$$

$$(4) \quad 2(y + 4) + 4(y + 1) =$$

$$(5) \quad 4(a + 4) + 4(a + 4) =$$

$$(6) \quad 3(x + 1) + 4(x - 1) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4(b + 2) + 4(b + 2) =$$

$$(2) \quad 3a + 4(4a + 1y) =$$

$$(3) \quad 4y + 3(-1y + 2x) =$$

$$(4) \quad 4(x + 4) + 4(x - 4) =$$

$$(5) \quad 4(x + 4) + 2(x - 2) =$$

$$(6) \quad 2a + 2(3a + 3y) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4a + 4(-1a + 4b) =$$

$$(2) \quad 2y + 3(-1y + 2b) =$$

$$(3) \quad 2y + 3(-3y + 2x) =$$

$$(4) \quad 3(b + 1) + 4(b + 3) =$$

$$(5) \quad 4(a + 1) + 2(a - 1) =$$

$$(6) \quad 4(a + 2) + 3(a + 5) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 3(y + 2) + 2(y + 3) =$$

$$(2) \quad 5(a + 4) + 4(a + 3) =$$

$$(3) \quad 4(a + 4) + 3(a + 1) =$$

$$(4) \quad 2(x + 3) + 2(x + 2) =$$

$$(5) \quad 4y + 3(-1y + 1a) =$$

$$(6) \quad 3(b + 4) + 2(b - 5) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (y + 4)^2 =$$

$$(3) \quad (x - 4)^2 =$$

$$(4) \quad (x - 3)^2 =$$

$$(5) \quad (a + 4)(a - 4) =$$

$$(6) \quad (a - 7)^2 =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (a + 6)^2 =$$

$$(3) \quad (x + 8)^2 =$$

$$(4) \quad (a + 7)^2 =$$

$$(5) \quad (a - 6)^2 =$$

$$(6) \quad (y - 2)^2 =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 5a + 3(2a + 1b) =$$

$$(2) \quad 3y + 3(3y + 1a) =$$

$$(3) \quad 5(x + 1) + 2(x - 4) =$$

$$(4) \quad 4y + 3(3y + 4x) =$$

$$(5) \quad 5(x + 1) + 4(x + 1) =$$

$$(6) \quad 3(a + 4) + 4(a + 5) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (y + 6)^2 =$$

$$(2) \quad (a + 4)^2 =$$

$$(3) \quad (y + 7)(y - 7) =$$

$$(4) \quad (y - 2)^2 =$$

$$(5) \quad (x - 4)^2 =$$

$$(6) \quad (a - 6)^2 =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (a - 6)^2 =$$

$$(2) \quad (x + 3)^2 =$$

$$(3) \quad (x + 6)^2 =$$

$$(4) \quad (y + 8)^2 =$$

$$(5) \quad (x + 8)(x - 8) =$$

$$(6) \quad (a - 5)^2 =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (y + 6)(y + 6) =$$

$$(2) \quad 4b(3b - 4) =$$

$$(3) \quad (x - 4)(x + 2) =$$

$$(4) \quad 4b(1b + 1) =$$

$$(5) \quad 3a(2a + 6) =$$

$$(6) \quad 3y(1y - 3x) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (y + 3)(y - 3) =$$

$$(2) \quad (x + 5)^2 =$$

$$(3) \quad (y - 2)^2 =$$

$$(4) \quad (y + 4)(y - 4) =$$

$$(5) \quad (x + 3)(x - 3) =$$

$$(6) \quad (x + 2)(x - 2) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 3b(3b + 6x) =$$

$$(2) \quad 5b(3b - 2y) =$$

$$(3) \quad 4x(1x - 3) =$$

$$(4) \quad 2x(3x + 3b) =$$

$$(5) \quad (x + 3)(x - 5) =$$

$$(6) \quad 4x(2x - 3a) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 4y(3y + 6) =$$

$$(2) \quad 4b(2b + 3y) =$$

$$(3) \quad (y - 3)(y + 1) =$$

$$(4) \quad 4b(3b - 4y) =$$

$$(5) \quad (y - 6)(y - 5) =$$

$$(6) \quad 2y(3y + 4x) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad (y - 1)(y - 2) =$$

$$(2) \quad (a + 2)(a + 5) =$$

$$(3) \quad 3a(1a - 1) =$$

$$(4) \quad 3a(2a + 6) =$$

$$(5) \quad 5x(3x - 3) =$$

$$(6) \quad (a - 3)(a + 5) =$$

Section 3. Factorisation

Factor each expression completely.

$$(1) \quad 3y(4y - 4a) =$$

$$(2) \quad 3x(2x + 4) =$$

$$(3) \quad (x + 4)(x + 6) =$$

$$(4) \quad 2y(1y + 3) =$$

$$(5) \quad (a - 2)(a + 5) =$$

$$(6) \quad (y + 6)(y + 3) =$$

Section 3 - Solutions

21a-1: $y^2 + 1y - 2$

21a-2: $y^2 - 6y + 8$

21a-3: $x^2 - 1$

21a-4: $6a^2 + 27a$

21a-5: $y^2 - 8y + 12$

21a-6: $x^2 - 6x + 9$

21b-1: $y^2 + 1y - 2$

21b-2: $5a^2 - 40a$

21b-3: $12x^2 + 36x$

21b-4: $y^2 + 2y - 8$

21b-5: $y^2 - 8y + 12$

21b-6: $x^2 - 6x + 9$

22a-1: $y^2 + 1y - 2$

22a-2: $5a^2 - 40a$

22a-3: $12x^2 + 36x$

Section 3 - Solutions (continued)

23b-1: $4x^2 + 14x$

23b-2: $5a^2 - 40a$

23b-3: $x^2 - 1$

23b-4: $6a^2 + 27a$

23b-5: $4y^2 + 36y$

23b-6: $8a^2 - 28a$

24a-1: $4x^2 + 14x$

24a-2: $5a^2 - 40a$

24a-3: $12x^2 + 36x$

24a-4: $y^2 + 2y - 8$

24a-5: $4y^2 + 36y$

24a-6: $x^2 - 6x + 9$

24b-1: $y^2 + 1y - 2$

24b-2: $5a^2 - 40a$

24b-3: $12x^2 + 36x$

Section 3 - Solutions (continued)

24b-4: $y^2 + 2y - 8$

24b-5: $4y^2 + 36y$

24b-6: $8a^2 - 28a$

25a-1: $y^2 + 1y - 2$

25a-2: $5a^2 - 40a$

25a-3: $x^2 - 1$

25a-4: $y^2 + 2y - 8$

25a-5: $4y^2 + 36y$

25a-6: $x^2 - 6x + 9$

25b-1: $4x^2 + 14x$

25b-2: $y^2 - 6y + 8$

25b-3: $12x^2 + 36x$

25b-4: $6a^2 + 27a$

25b-5: $4y^2 + 36y$

25b-6: $8a^2 - 28a$

Section 3 - Solutions (continued)

22a-4: $y^2 + 2y - 8$

22a-5: $4y^2 + 36y$

22a-6: $x^2 - 6x + 9$

22b-1: $4x^2 + 14x$

22b-2: $5a^2 - 40a$

22b-3: $12x^2 + 36x$

22b-4: $y^2 + 2y - 8$

22b-5: $y^2 - 8y + 12$

22b-6: $x^2 - 6x + 9$

23a-1: $y^2 + 1y - 2$

23a-2: $5a^2 - 40a$

23a-3: $x^2 - 1$

23a-4: $y^2 + 2y - 8$

23a-5: $4y^2 + 36y$

23a-6: $x^2 - 6x + 9$

Section 3 - Solutions (continued)

26a-1: $y^2 + 1y - 2$

26a-2: $y^2 - 6y + 8$

26a-3: $x^2 - 1$

26a-4: $6a^2 + 27a$

26a-5: $y^2 - 8y + 12$

26a-6: $8a^2 - 28a$

26b-1: $y^2 + 1y - 2$

26b-2: $y^2 - 6y + 8$

26b-3: $12x^2 + 36x$

26b-4: $y^2 + 2y - 8$

26b-5: $y^2 - 8y + 12$

26b-6: $8a^2 - 28a$

27a-1: $4x^2 + 14x$

27a-2: $5a^2 - 40a$

27a-3: $x^2 - 1$

Section 3 - Solutions (continued)

28b-1: $4x^2 + 14x$

28b-2: $y^2 - 6y + 8$

28b-3: $12x^2 + 36x$

28b-4: $6a^2 + 27a$

28b-5: $4y^2 + 36y$

28b-6: $x^2 - 6x + 9$

29a-1: $4x^2 + 14x$

29a-2: $5a^2 - 40a$

29a-3: $12x^2 + 36x$

29a-4: $y^2 + 2y - 8$

29a-5: $4y^2 + 36y$

29a-6: $x^2 - 6x + 9$

29b-1: $y^2 + 1y - 2$

29b-2: $5a^2 - 40a$

29b-3: $12x^2 + 36x$

Section 3 - Solutions (continued)

29b-4: $6a^2 + 27a$

29b-5: $4y^2 + 36y$

29b-6: $x^2 - 6x + 9$

30a-1: $y^2 + 1y - 2$

30a-2: $y^2 - 6y + 8$

30a-3: $12x^2 + 36x$

30a-4: $y^2 + 2y - 8$

30a-5: $4y^2 + 36y$

30a-6: $x^2 - 6x + 9$

30b-1: $4x^2 + 14x$

30b-2: $y^2 - 6y + 8$

30b-3: $x^2 - 1$

30b-4: $6a^2 + 27a$

30b-5: $4y^2 + 36y$

30b-6: $x^2 - 6x + 9$

Section 3 - Solutions (continued)

27a-4: $6a^2 + 27a$

27a-5: $y^2 - 8y + 12$

27a-6: $8a^2 - 28a$

27b-1: $y^2 + 1y - 2$

27b-2: $5a^2 - 40a$

27b-3: $x^2 - 1$

27b-4: $y^2 + 2y - 8$

27b-5: $4y^2 + 36y$

27b-6: $x^2 - 6x + 9$

28a-1: $y^2 + 1y - 2$

28a-2: $y^2 - 6y + 8$

28a-3: $x^2 - 1$

28a-4: $y^2 + 2y - 8$

28a-5: $4y^2 + 36y$

28a-6: $8a^2 - 28a$